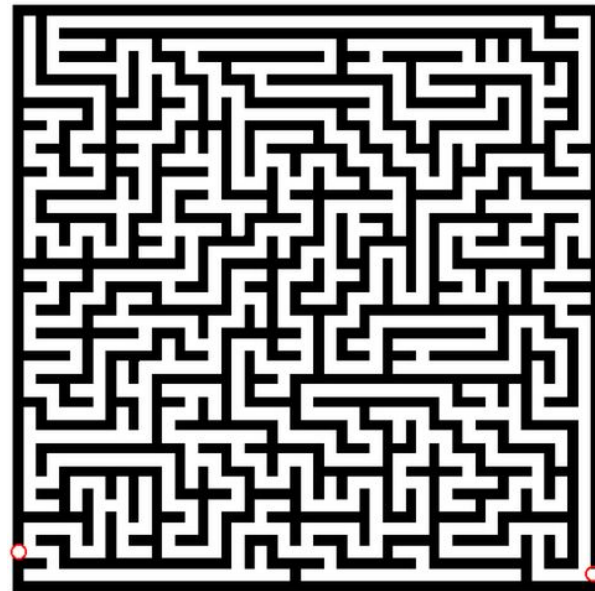


LECTURE: 5

PROBLEM SPACE AND SEARCH

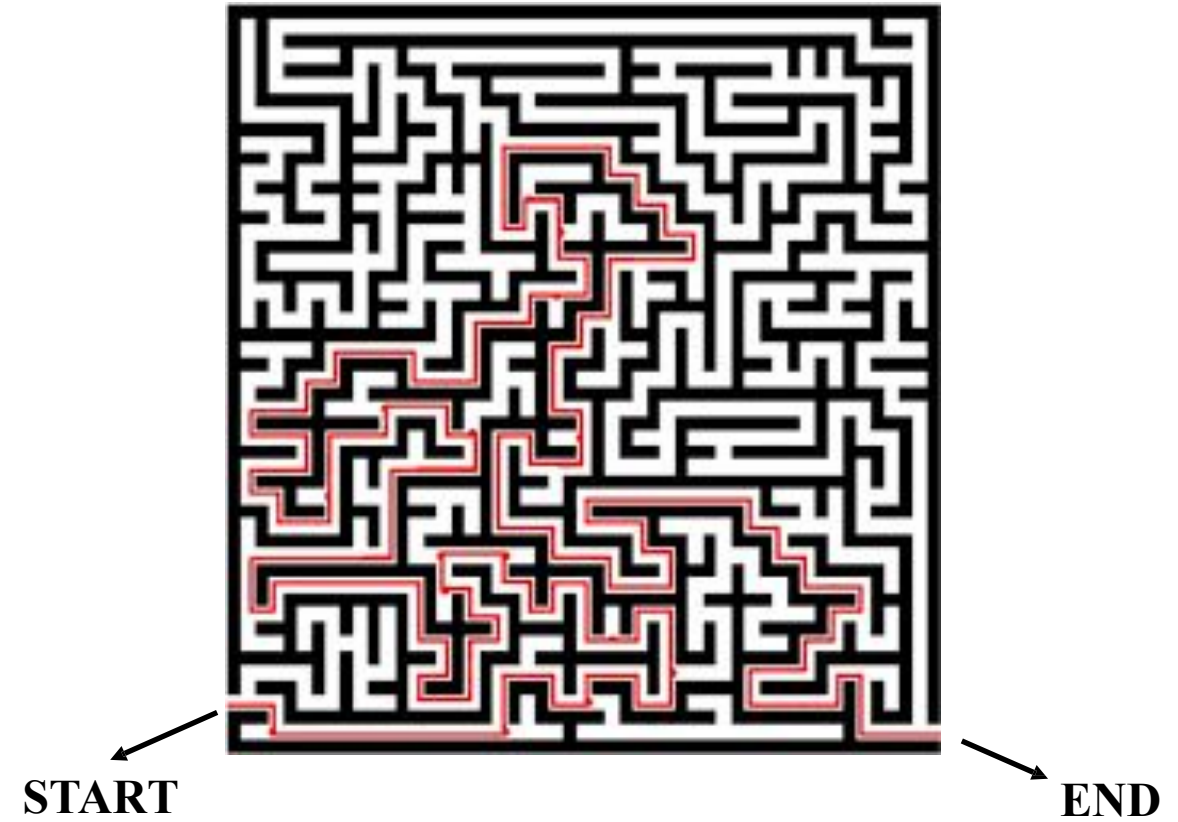
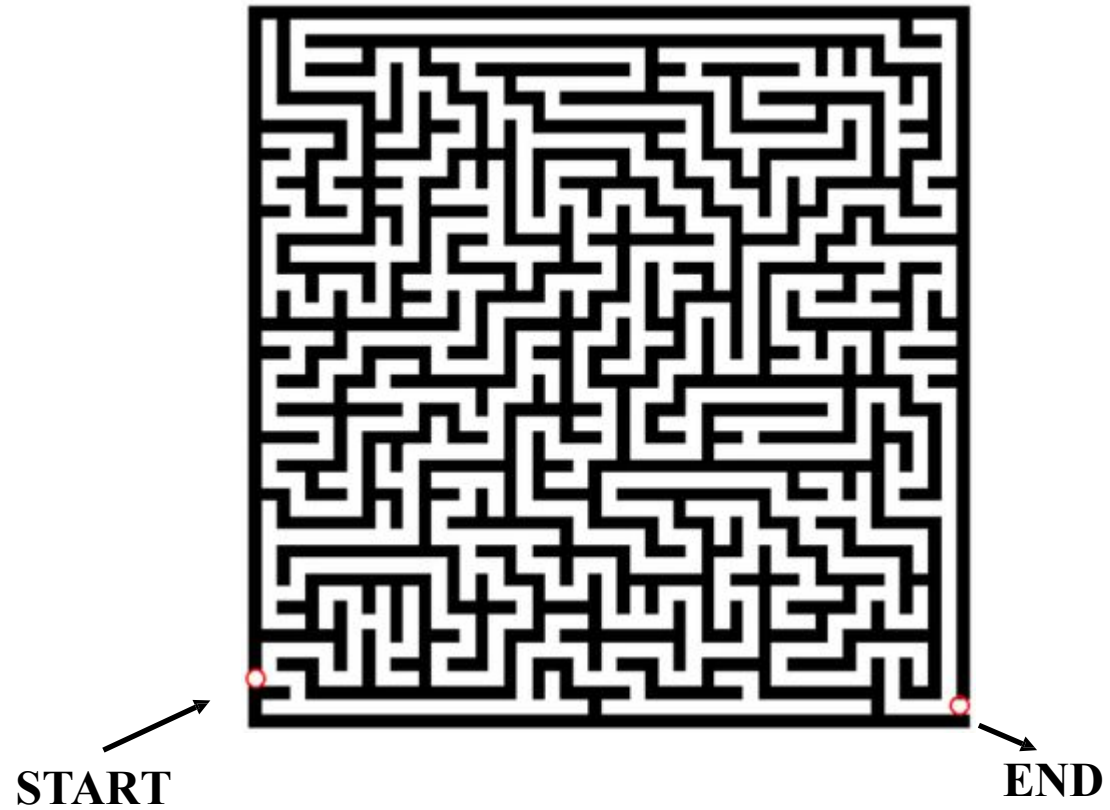


How will you solve this?

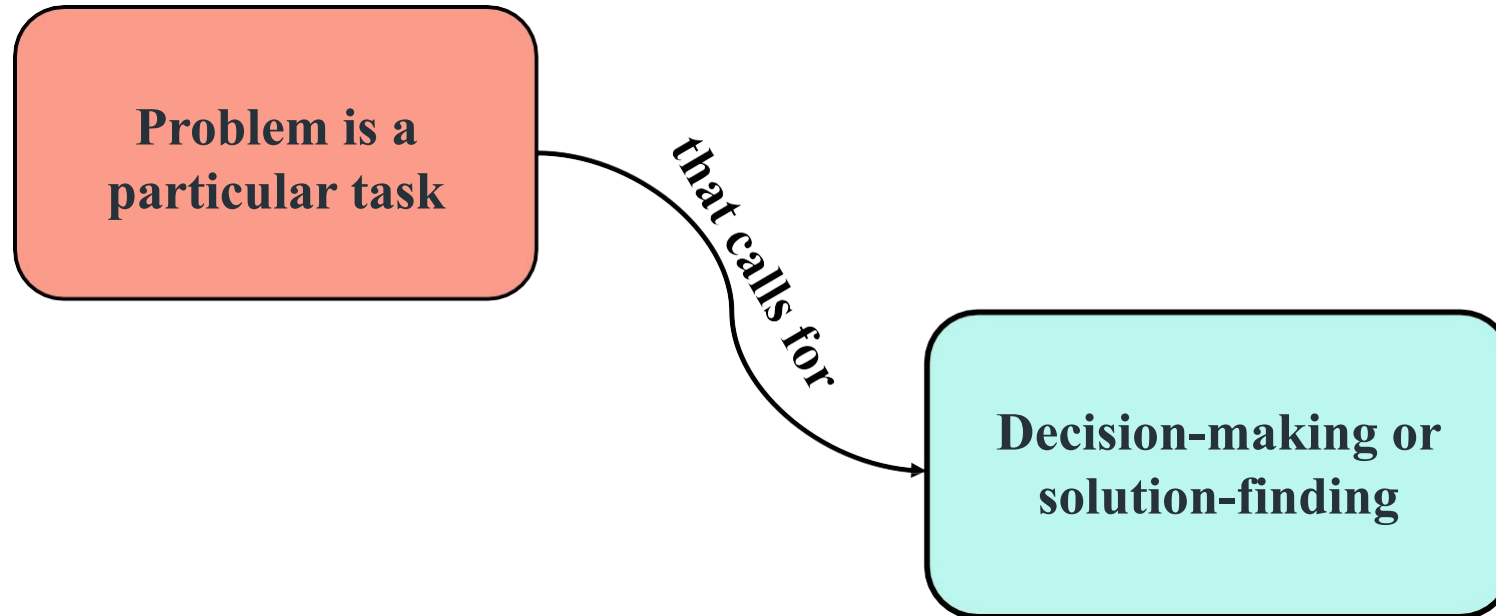




SOLVED:



What are problems in AI?





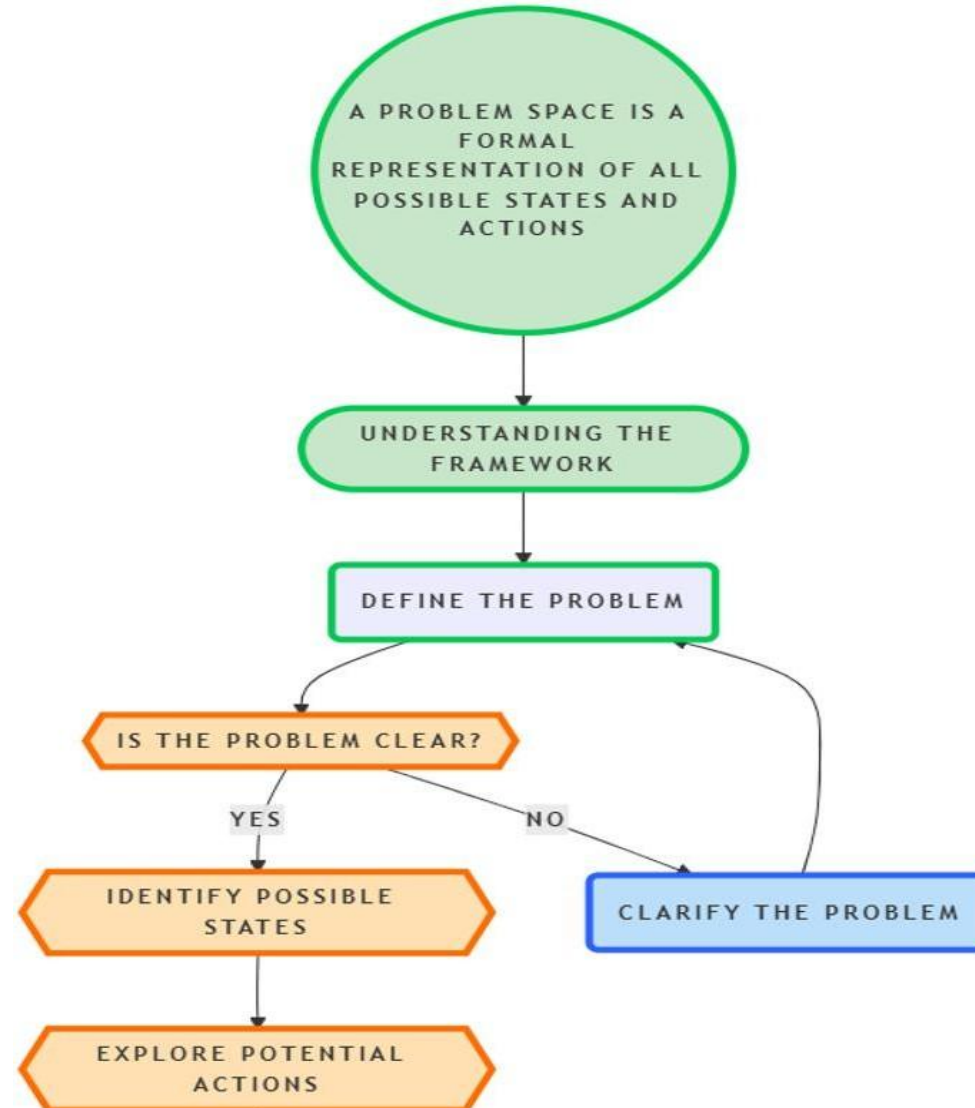
Components of Problems in AI:

- **Initial State:** The problem's starting condition.
- **Goal State:** The desired outcome or solution.
- **Operators:** Actions to transition between states.
- **Restrictions:** Rules or constraints for solving the problem.

Example: Chess Game

- **Initial State:** Starting positions of all chess pieces.
- **Objective State:** Achieving checkmate.
- **Operators:** Permissible moves for each piece.
- **Constraints:** Rules governing chess gameplay.

Problem Space:





Problem Formulation by State Space Search

1. State space formally defines problem structure.
2. Graph (V, E) represents state space.
3. Nodes (V) correspond to problem states.
4. Edges (E) represent applicable actions.
5. Directed arcs connect parent and child nodes.
6. Parent precedes, child succeeds in graph.

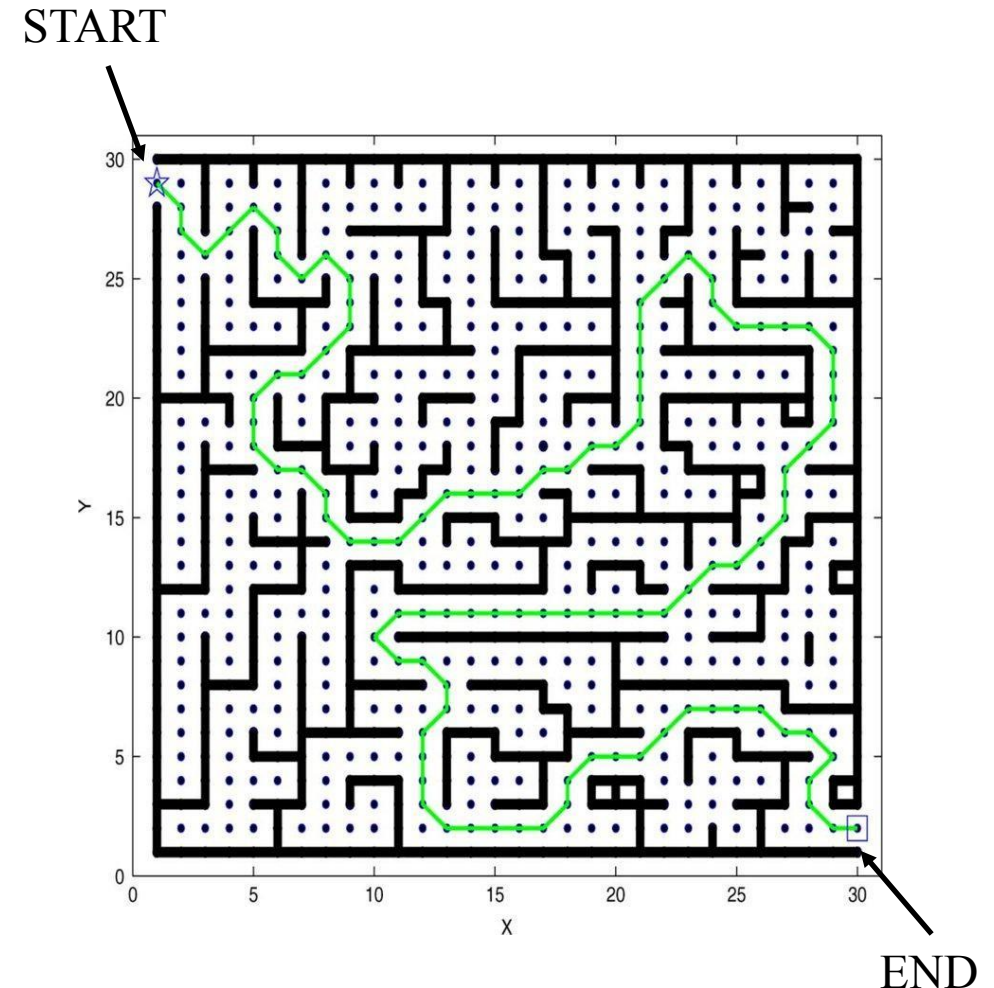


Components of Problem Spaces:

- **States:** Possible scenarios within the problem.
- **State Space:** All reachable states from start.
- **Paths:** Connections between states via operators.

Example 1: Maze-Solving Problem

- **State Space:** The maze structure itself.
- **States:** All possible positions in the maze.
- **Paths:** Routes connecting the start to the exit.



Example 2—Puzzle Solving:

1		2
3	4	5
6	7	7

Current State

1	4	2
3	7	5
6		7

Target State

